

Module 1 Cheatsheet: Data Analytics and Generative AI

Important Terms

Term	Description
Generative AI	Is a category of AI that focuses on creating new, synthetic data. Unlike traditional AI models that predict or classify, generative models generate entirely new data points, opening a realm of possibilities for data analytics.
Data Augmentation	Is a powerful technique for improving the performance of machine learning models, especially when the training data sets of the models are small or imbalanced.
Data Preparation	Is crucial in the data analytics journey. To prepare raw data for analysis, it must be cleaned, transformed, and arranged in a format that makes it easy for analytical tools to use.
Database Querying	Is the process of working with a database to extract relevant details for analysis. It includes interacting with a database and retrieving data that satisfies certain criteria using query languages, most often SQL (Structured Query Language).

Term	Description
Data Q&A	Asking questions and getting answers about certain data sets or data analysis activities is called Q&A for data. With Q&A, you can accomplish data exploration, extract insights, and get a better comprehension of the underlying patterns and trends in the data.

Generative AI Platforms/Tools Used in this Module

Task Performed	Generative AI Platform/Tools
Data Augmentation and Generation	DataRobot  Colab  ChatGPT  Bard  MOSTLY.AI  universaldata 
Data Preparation	ChatCSV  Tomat.AI 
Data Querying	SQLthroughAI  dbsensei 
Data Insights through QnA	Akkio 

Some Generic Prompts

Task	Prompt	Example
Create dataset in a particular domain	Create <.....> dataset for <.....>	Create patient data set for the symptoms of diabetes
Create dataset with specific attributes and format	Create a dataset with attributes as <.....> in a <.....> format	Create a dataset with attributes as temperature in Fahrenheit, Temperature Category, Humidity in percentage, Rain and Snow as categorical type in yes or no categories, Month, Year in a CSV format
Get the insights through Q&A: Finding highest value within the data attribute	Identify the <.....> with the highest <.....>	Identify the products with the highest sales What are the top selling products?
Get the insights through Q&A: See patterns of a data attribute over a period of time	How has <.....> changed over time?	How has the quantity ordered changed over time?
Identify missing data	Write a <...> code to Identify <.....> missing values. Identify the attributes with missing data	Write a <i>python</i> code to Identify <i>the columns</i> with missing values (ChatGPT) Identify the attributes with missing data (ChatCSV)

Task	Prompt	Example
Handling missing values	Write a <...> code to replace missing values with <.....> in the dataset Replace the missing values <...> in the <...> and save the updated dataset	Write a <i>python</i> code to replace missing values with <i>mean values</i> in the dataset (ChatGPT) Replace the missing values with <i>the mean value</i> in the <i>Screen_size_cm</i> column and save the updated dataset (ChatCSV)
Join two tables	Write a SQL query to join <.....> with <.....> on the <.....> as a primary key Join <.....> with <.....> on <.....> as primary key	Write an SQL query to join <i>customer table</i> with the <i>product sales tables</i> on <i>product ID</i> as primary key (ChatGPT) Join <i>customer table</i> with the <i>product sales table</i> on <i>product ID</i> as primary key (dbsensei)
Create database	Write a <.....> query to Create a database on <.....>, create a <....> and insert values in these tables Create a database on <.....>, create tables, insert values	Write a SQL query to Create a database on <i>sales</i>, create a <i>customer table</i>, a <i>sales table</i>, a <i>product table</i> and insert values in these tables (ChatGPT) Create a database on <i>customers and sales</i>, create tables, insert values (dbsensei)

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